Exp No:12

Experiment to detect outliers in a given data set.

Program:

import numpy as np

array=np.random.randint(1,100,16)

array

Output:

```
import numpy as np
array=np.random.randint(1,100,9)
array
```

```
array([73, 62, 71, 78, 48, 34, 79, 58,  2])
```

```
array.mean()
```

```
np.float64(56.111111111111114)
```

```
np.percentile(array,25)
```

```
np.float64(48.0)
```

```
np.percentile(array,50)
```

```
np.float64(62.0)
```

```
np.percentile(array,75)
```

```
np.float64(73.0)
```

```
np.percentile(array,100)
```

```
np.float64(79.0)
```

Program:

```
def outDetection(array):
sorted(array)
Q1,Q3=np.percentile(array,[25,75])
IQR=Q3-Q1
lr=Q1-(1.5*IQR)
ur=Q3+(1.5*IQR)
return lr,ur
lr,ur=outDetection(array)
lr,ur
```

Output:

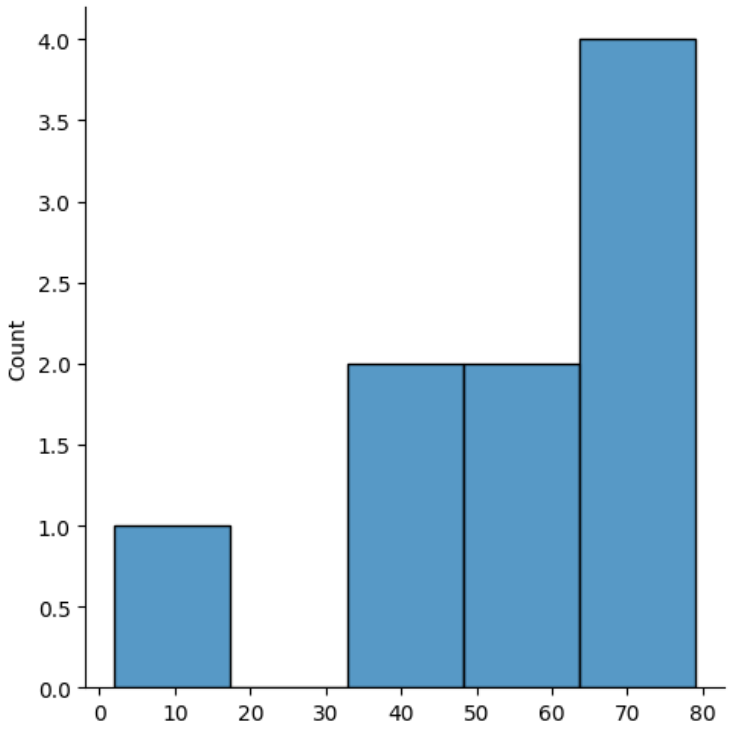
```
def outDetection(array):
  sorted(array)
  Q1,Q3=np.percentile(array,[25,75])
  IQR=Q3-Q1
  lr=Q1-(1.5*IQR)
  ur=Q3+(1.5*IQR)
  return lr,ur

lr,ur=outDetection(array)
lr,ur
```

```
(np.float64(10.5), np.float64(110.5))
```

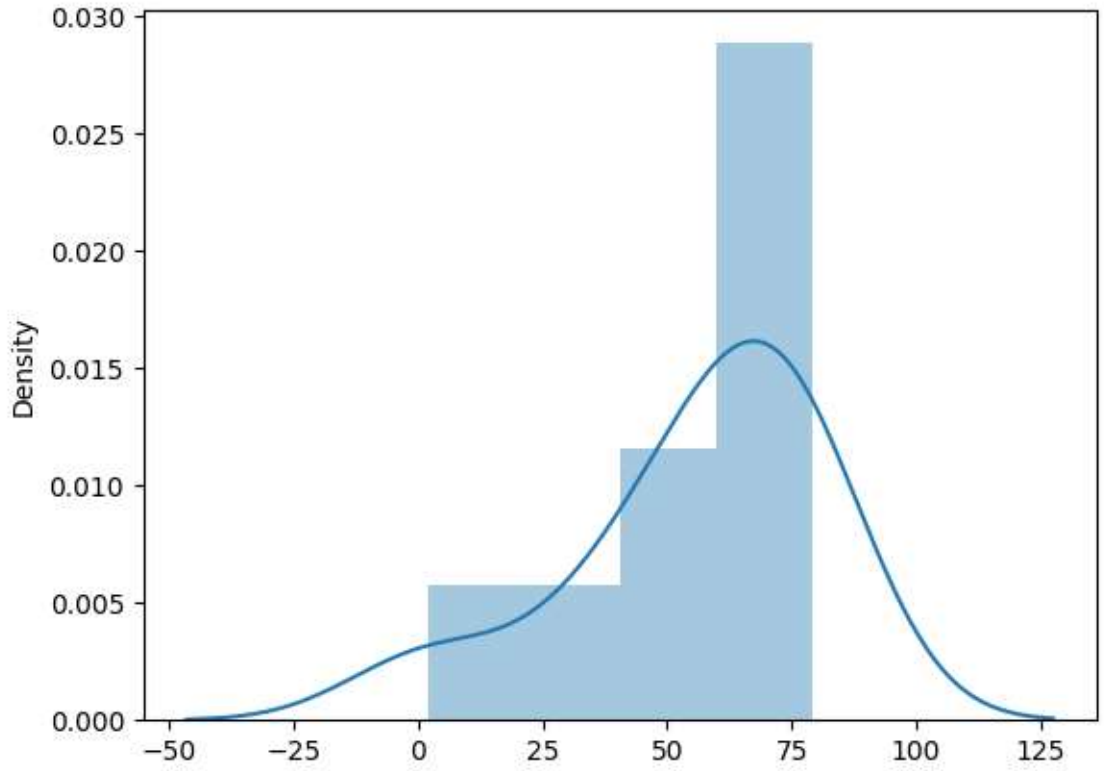
Program:

```
import seaborn as sns
%matplotlib inline
sns.displot(array)
```



sns.distplot(array)

Output:

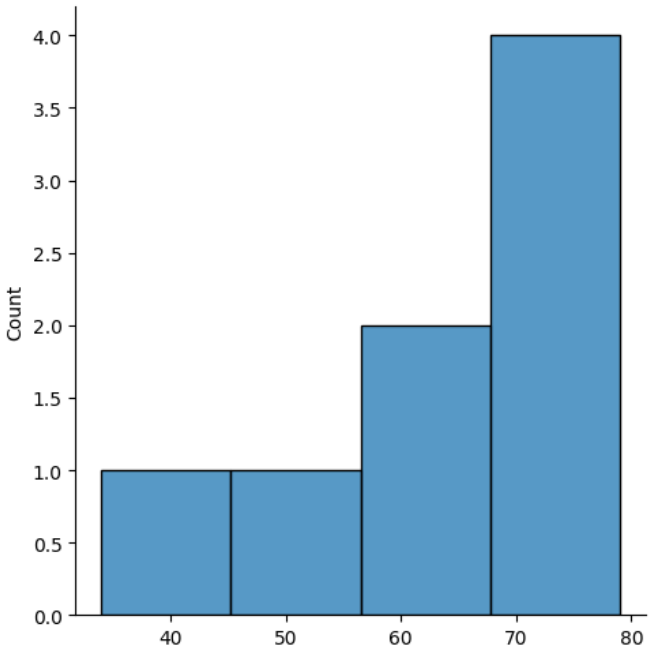


new_array=array[(array>lr) & (array<ur)]

new_array

sns.displot(new_array)

Output:



sns.distplot(final_array)

